

Simulation Depth & Precision Module (SDPM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Governed Space: Simulation Depth & Precision

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Canonical Language: English (UCL™)

Governed Space

Simulation Depth & Precision

Canonical Definition

Simulation Depth & Precision Module (SDPM) defines the governance conditions under which the required analytical depth, representational precision, temporal and spatial granularity, parameter resolution, interaction detail, scenario intensity, uncertainty treatment, and consequence-proportionate simulation rigor are established for a defined Simulation Purpose, Scope, Exclusions, and Applicability Context.

SDPM determines how deeply and precisely the governed simulation inquiry must examine the Simulation Object without prescribing a particular simulation model, technology, numerical method, software platform, or validation methodology.

It prevents a simulation from being treated as sufficiently informative merely because the correct object and scope were selected.

Operational Role

SDPM serves as the fifth and final internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The preceding modules establish:

SPM — Why is the simulation being performed?

SSCM — What must the simulation cover?

SEXM — What remains outside the governed Simulation Scope?

SACM — Within which contexts is the simulation inquiry intended to apply?

SDPM asks:

At what depth and precision must the simulation inquiry operate to be adequate for its intended purpose and consequence?

The governed progression is:

Simulation Purpose → Simulation Scope → Simulation Exclusions → Simulation Applicability Context → Required Simulation Depth & Precision

SDPM therefore completes the simulation inquiry definition before the architecture proceeds into governance of how reality is represented inside the simulation.

Module Operational Space

SDPM governs: Simulation Depth, Simulation Precision, required simulation rigor, inquiry granularity, abstraction-depth requirements, parameter precision, temporal granularity, spatial granularity, behavioral detail, interaction detail, dependency detail, state resolution, event resolution, scenario intensity, scenario granularity, edge-condition depth, failure-condition depth, uncertainty-treatment depth, sensitivity-analysis depth, consequence-proportionate rigor, depth differentiation, precision differentiation, minimum acceptable depth, minimum acceptable precision, depth sufficiency, precision sufficiency, depth limitations, precision limitations, depth escalation, precision escalation, depth reduction, precision reduction, depth change, precision change, and Simulation Depth & Precision traceability.

Module Function

A simulation may have: the correct Simulation Object, a legitimate Simulation Purpose, a properly bounded Simulation Scope, explicit exclusions, and an appropriate Applicability Context,

while still being incapable of answering the simulation question with sufficient depth.

For example:

Purpose: Assess collision behavior.

Scope: Vehicle frontal collision.

Applicability: Defined vehicle configuration and impact conditions.

But the simulation represents the event only at a level sufficient for

approximate structural deformation.

That simulation may be appropriate for early design exploration while being insufficient for detailed occupant-injury analysis.

SDPM prevents:

Correct Simulation Question → Insufficient Simulation Depth → Apparently Precise Output → Overstated Simulation Meaning

Its function is therefore to establish the required level of examination before apparent numerical precision is mistaken for methodological sufficiency.

Simulation Depth

Simulation Depth defines how extensively the simulation inquiry must examine the relevant characteristics, relationships, behaviors, interactions, dependencies, states, scenarios, and consequences within the governed Simulation Scope.

Depth answers:

How far must the simulation inquiry go?

Simulation Depth may determine whether the inquiry requires examination of: primary behavior only, secondary effects, interaction effects, cascading effects, dependencies, failure modes, edge conditions, rare conditions, recovery behavior, long-duration effects, or consequence propagation.

Simulation Precision

Simulation Precision defines the required degree of differentiation and granularity with which simulation-relevant variables, states, relationships, events, parameters, and outputs must be represented for the Simulation Purpose.

Precision answers:

How finely must the simulation distinguish simulation-relevant differences?

Precision may involve: parameter resolution, spatial resolution, temporal resolution, state resolution, behavioral resolution, event resolution, measurement granularity, interaction granularity, or output granularity.

Depth ≠ Precision

Depth and precision are related but not interchangeable.

A simulation may be:

deep but low-precision

or:

narrow but high-precision.

For example, a simulation may examine many interacting system dependencies while representing each through coarse parameter ranges.

Another simulation may represent one narrowly bounded physical process at extremely fine numerical resolution.

SDPM therefore governs both dimensions independently where necessary.
Precision $\neq$ Accuracy

SDPM establishes an important distinction:

Simulation Precision does not itself establish Simulation Accuracy.

A simulation may produce highly precise numerical outputs from an inadequate model, incorrect assumptions, unsuitable inputs, or insufficient reality representation.

Therefore:

More decimal places $\neq$ Better simulation.

Precision must be justified by the simulation inquiry. Depth $\neq$ Fidelity

Simulation Depth governs how extensively the inquiry must examine the problem.

Representation Fidelity, governed subsequently within the Reality Representation Standard (RRS), concerns how faithfully selected aspects of reality are represented inside the simulation.

A simulation may therefore require high analytical depth while legitimately using different fidelity levels across different represented elements.

Purpose-Proportionate Depth

Required Simulation Depth should derive from the Simulation Purpose.

A simulation intended for:

early conceptual exploration

may legitimately require less depth than one intended for:

engineering qualification,

safety-critical analysis,

or:

high-consequence operational decision support.

SDPM does not establish one universal depth requirement.

It establishes that depth must be purpose-proportionate.

Consequence-Proportionate Rigor

Where simulation outputs may influence high-consequence decisions, the required depth and precision may need to increase.

Conceptually:

Higher Consequence → Stronger Justification for Required Simulation Rigor

This does not mean every high-consequence simulation must always maximize every form of precision.

It means that reductions in depth or precision require stronger justification where their consequences are material.

Minimum Sufficient Depth

The objective of SDPM is not maximum simulation complexity.

The objective is:

sufficient simulation depth for the governed purpose.

Unnecessary depth may: increase computational cost, increase modeling complexity, introduce additional assumptions, increase data requirements, obscure material relationships, and reduce operational usability.

SDPM therefore supports the minimum sufficient depth principle.

Minimum Sufficient Precision

Likewise, simulation precision should be sufficient to distinguish differences that materially affect the simulation inquiry.

Precision beyond what the purpose can meaningfully use may create: false confidence, unnecessary computational burden, misleading apparent certainty, or complexity without additional decision value.

The governing question is not:

How precise can the simulation become?

It is:

How precise must the simulation be for the intended purpose?

Depth Dimensions

Simulation Depth may be decomposed into relevant dimensions.

These may include: Structural Depth

How extensively must internal structures and relationships be examined?

Behavioral Depth

How extensively must behavior, behavioral variation, and behavioral transitions be represented?

Interaction Depth

How extensively must interactions among simulated entities be examined?

Dependency Depth

How deeply must external and internal dependencies be represented?

Failure Depth

Which failure conditions, degraded states, and failure combinations must be examined?

Consequence Depth

How far must downstream effects and consequence propagation be followed?

Scenario Depth

How broad and intensive must scenario exploration become?

Temporal Depth

How far through time must simulation behavior be examined?

Different dimensions may legitimately require different levels of depth.

Precision Dimensions

Precision may similarly be multidimensional.

Relevant dimensions may include:

Spatial Precision

Temporal Precision

Parameter Precision

State Precision

Behavioral Precision

Interaction Precision

Event Precision

Output Precision

A simulation does not need identical precision across all dimensions.

Differentiated Depth Rule

SDPM permits different components of the same simulation inquiry to operate at different depths where justified.

For example:

Primary safety mechanism → High Depth

Secondary cosmetic behavior → Low Depth

This allows simulation resources to concentrate on what materially affects the Simulation Purpose.

Differentiated Precision Rule

Different simulation variables may require different precision.

For example:

Collision timing → millisecond-level resolution

while:

ambient temperature → broader interval representation

may be sufficient for the same simulation.

Precision should follow materiality rather than uniformity.

Materiality Principle

A simulation characteristic is depth- or precision-material where insufficient treatment could materially alter: simulation behavior, simulation outputs, interpretation, decision relevance, scenario outcome, consequence assessment, or downstream reliance.

Material dimensions should receive explicit depth and precision requirements.

Temporal Granularity

Temporal granularity determines how finely simulation time must be represented.

Insufficient temporal granularity may conceal: rapid transitions, control instability, collisions, short-duration failures, timing dependencies, cascading events, or synchronization problems.

Excessive temporal granularity may create unnecessary computational burden

without improving the governed inquiry.

Spatial Granularity

Spatial granularity determines how finely physical or virtual space must be represented.

The required level depends upon the simulation purpose.

A regional traffic-flow simulation and a vehicle collision simulation legitimately require radically different spatial precision.

Parameter Precision

Simulation parameters should have precision appropriate to their material influence.

Parameter precision should not exceed the defensible precision of the underlying information merely to make the simulation appear more exact.

Input Precision Constraint

Where an input is known only within a broad range, the simulation should not silently convert that uncertainty into artificial precision.

Conceptually:

Input Knowledge Precision → Limits Defensible Simulation Precision

unless additional modeling explicitly governs the uncertainty.

State Resolution

Simulation states should be differentiated sufficiently to capture state transitions material to the inquiry.

For example:

Operational / Failed

may be sufficient for one simulation.

Another may require:

Nominal → Degraded → Critical → Failed → Recovering → Restored

SDPM governs whether the chosen resolution is sufficient for the intended inquiry.

Interaction Resolution

Where outcomes depend upon interactions, the simulation must distinguish interaction characteristics at sufficient depth.

This may include: timing, sequencing, dependency, coordination, conflict, feedback, communication, or emergent interaction effects.

Scenario Depth

Scenario Depth establishes how extensively the simulation inquiry must explore the relevant scenario space.

This may include:

Nominal Conditions

Expected Variations

Edge Conditions

Failure Conditions

Rare Events

Adversarial Conditions

Compound Conditions

The required scenario depth derives from the Simulation Purpose and consequence context.

Detailed Scenario Design itself remains governed by the dedicated downstream SIMULOS® scenario governance space.

Edge-Condition Depth

Where boundary conditions materially influence simulation meaning, SDPM may require explicit examination near or beyond expected operating boundaries.

A simulation intended for robust safety analysis should not necessarily stop at average conditions.

Failure-Condition Depth

Failure-condition depth establishes how extensively failure modes and combinations must be examined.

Possible levels may include:

Single Failure

Multiple Independent Failures

Dependent Failures

Cascading Failures

Recovery Failure

The appropriate level depends upon purpose and consequence.

Uncertainty-Treatment Depth

Simulation Depth may require uncertainty to be examined rather than represented as a single deterministic value.

This may involve: ranges, distributions, sensitivity, confidence bounds, scenario variation, parameter uncertainty, or structural uncertainty.

SDPM establishes the required depth of uncertainty treatment.

It does not replace specialized validation or integrity governance.

Sensitivity Depth

Where simulation outcomes are highly sensitive to particular assumptions or parameters, greater depth may be required around those variables.

A parameter with negligible effect does not necessarily require the same precision as one capable of reversing the simulation conclusion.

Depth Sufficiency

Simulation Depth Sufficiency asks:

Is the established depth sufficient to address the Simulation Purpose within the governed Scope and Applicability Context?

Possible outcomes may include:

Sufficient

Conditionally Sufficient

Partially Sufficient

Insufficient

or:

Undetermined Precision Sufficiency

Simulation Precision Sufficiency asks:

Is the established precision sufficient to distinguish the simulation-relevant differences material to the governed inquiry?

Possible outcomes may include:

Sufficient

Conditionally Sufficient

Partially Sufficient

Insufficient

or:

Undetermined Depth–Precision Alignment

Depth and precision should remain mutually coherent.

A simulation should not use extreme numerical precision to disguise shallow analytical coverage.

Likewise, extensive analytical coverage should not rely upon resolution too coarse to distinguish material behavior.

SDPM therefore supports:

Required Depth ↔ Required Precision ↔ Simulation Purpose Depth–Purpose Alignment

Every material depth requirement should be traceable to the Simulation Purpose, Scope, Applicability Context, or consequence.

This prevents complexity from becoming arbitrary. Precision–Purpose Alignment

Precision should likewise remain connected to the actual simulation question.

A precision requirement without a meaningful simulation purpose may create computational cost without methodological value.

Depth Limitation

Where the required Simulation Depth cannot be achieved, the limitation should be explicit.

The simulation may still proceed where appropriate, but downstream interpretation must remain bounded by the achieved depth.

Precision Limitation

Where required precision cannot be achieved, the resulting limitation should remain visible.

The simulation should not silently present lower-resolution analysis as satisfying a higher-resolution requirement.

No False Precision Rule

Simulation outputs must not imply a level of precision materially stronger than the governed simulation basis can support.

No Complexity Substitution Rule

Greater model complexity must not be treated as automatic evidence of sufficient Simulation Depth.

Complexity and methodological adequacy are not equivalent.

No Computational-Scale Substitution Rule

A simulation does not become methodologically stronger merely because it uses: more compute, more parameters, more iterations, larger models, more agents, or longer execution.

Computational scale must serve the simulation inquiry.

No Uniform Precision Rule

A simulation need not apply maximum or identical precision to every simulated element.

Precision should be allocated according to materiality.

No Depth Inflation Rule

A simulation must not claim deeper analytical coverage than was actually established and executed.

No Precision Inflation Rule

A simulation must not represent outputs as more precise than the underlying simulation conditions legitimately support.

No Purpose Transfer Rule

Depth sufficient for one Simulation Purpose must not automatically be considered sufficient for another purpose.

For example:

Depth sufficient for concept exploration

does not automatically become:

Depth sufficient for operational safety authorization.

Depth Escalation

Required depth may increase when: Simulation Purpose changes, consequence increases, new failure modes emerge, new dependencies become material, new operational environments are introduced, new autonomy is added, new interactions emerge, uncertainty increases, or downstream use becomes more demanding.

Precision Escalation

Required precision may increase where: smaller differences become decision-relevant, timing becomes more critical, spatial relationships become more consequential, new thresholds emerge, control behavior changes, or output interpretation requires finer differentiation.

Depth Reduction

Depth may legitimately be reduced where: the Simulation Purpose narrows, consequence decreases, irrelevant dimensions are removed, sensitivity analysis demonstrates low materiality, or a previously complex inquiry becomes bounded to a simpler question.

Reduction should be justified rather than assumed.

Precision Reduction

Precision may also be reduced where additional granularity produces no material improvement to the governed simulation inquiry.

This supports computational and methodological proportionality.

Simulation Resource Proportionality

SDPM recognizes that depth and precision consume resources.

These may include: computational capacity, energy, time, data, engineering effort, expert review, infrastructure, and financial resources.

The objective is not unlimited consumption.

The objective is:

sufficient governed simulation capability proportional to the purpose and consequence.

Resource-efficiency governance may be addressed by complementary OOF® architectures where applicable.

Depth & Precision Change

Material changes to: Simulation Purpose, Simulation Scope, exclusions, Applicability Context, Simulation Object, consequence, operational environment, or intended downstream use

may require reassessment of Simulation Depth & Precision.

Depth & Precision Traceability

A reviewer should be capable of determining:

Why was this depth selected?

Why was this precision selected?

Which dimensions received greater or lower treatment?

Which limitations remained?

and:

Was the achieved depth and precision sufficient for the original Simulation Purpose?

Simulation Depth & Precision Record

A governed record may preserve: Simulation Purpose, Simulation Scope, exclusions, Applicability Context, consequence context, material simulation dimensions, required depth by dimension, required precision by dimension, temporal granularity, spatial granularity, parameter precision, state resolution, interaction resolution, scenario depth, uncertainty-treatment depth, sensitivity requirements, depth limitations, precision limitations, sufficiency status, change triggers, and traceability.

Failure Conditions

SDPM governance failure may exist where: required Simulation Depth is undefined, required precision is undefined where materially necessary, depth does not correspond to Simulation Purpose, precision does not correspond to material simulation differences, high-consequence use relies upon unjustifiably shallow simulation, false numerical precision is presented, insufficient input precision is hidden, model complexity is substituted for methodological depth, computational scale is treated as evidence of adequacy, material failure conditions are omitted, material interactions are represented too coarsely, scenario depth is insufficient for the stated purpose, uncertainty treatment is materially insufficient, depth limitations are hidden, precision limitations are hidden, achieved depth is represented as stronger than it is, or simulation outputs are used for a purpose requiring materially greater depth or precision than was established.

Minimum Implementation Framework

1. Identify Material Depth & Precision Dimensions

Determine which structural, behavioral, temporal, spatial, parameter, interaction, scenario, failure, consequence, and uncertainty dimensions materially affect the Simulation Purpose. 2. Establish Required Simulation Depth

Define how extensively each material dimension must be examined. 3. Establish Required Simulation Precision

Define the granularity and resolution necessary to distinguish simulation-relevant differences. 4. Assess Sufficiency and Limitations

Determine whether the planned or achieved depth and precision are sufficient and explicitly preserve material limitations. 5. Preserve Traceability and Change Control

Maintain the Simulation Depth & Precision Record and reassess requirements when purpose, scope, applicability, consequence, or other material simulation

conditions change.

Governance Outputs

SDPM may produce: Simulation Depth Requirement, Simulation Precision Requirement, Simulation Depth & Precision Record, Simulation Rigor Profile, Depth Dimension Map, Precision Dimension Map, Structural Depth Requirement, Behavioral Depth Requirement, Interaction Depth Requirement, Dependency Depth Requirement, Failure Depth Requirement, Consequence Depth Requirement, Scenario Depth Requirement, Temporal Depth Requirement, Spatial Precision Requirement, Temporal Precision Requirement, Parameter Precision Requirement, State Resolution Requirement, Interaction Resolution Requirement, Event Resolution Requirement, Output Precision Requirement, Uncertainty-Treatment Requirement, Sensitivity Depth Requirement, Minimum Sufficient Depth Record, Minimum Sufficient Precision Record, Depth Sufficiency Assessment, Precision Sufficiency Assessment, Depth–Precision Alignment Record, Depth Limitation Record, Precision Limitation Record, Depth Escalation Record, Precision Escalation Record, Depth Reduction Record, Precision Reduction Record, Simulation Rigor Change Record, and Simulation Depth & Precision Traceability Map.

Use Case 1 — Autonomous Robot Safety Simulation

Scenario

An autonomous industrial robot is simulated before deployment around human workers.

The Simulation Purpose is to examine collision avoidance and emergency stopping.

A coarse simulation is sufficient to examine general navigation efficiency but not the timing and distance relationships relevant to human safety.

Application

SDPM identifies safety-critical dimensions requiring greater depth and precision:

human–robot interaction,

sensor-to-action latency,

stopping distance,

human movement variation,

failure response,

and:

emergency-state transitions.

The simulation may retain lower precision for dimensions that do not materially affect the safety inquiry. Result

Simulation resources are concentrated on the dimensions capable of changing the safety conclusion.

The simulation does not need maximum fidelity everywhere, but it must achieve sufficient depth and precision where the Simulation Purpose materially depends upon them. Use Case 2 — Spacecraft Mission Simulation

Scenario

A spacecraft mission simulation is used first for early mission architecture exploration and later proposed as part of a higher-consequence mission-planning process.

The original simulation uses broad parameter ranges and relatively coarse temporal resolution. Application

SDPM identifies that the new Simulation Purpose requires increased depth in:

propulsion behavior,

orbital interactions,

failure conditions,

navigation dependencies,

and:

mission-phase transitions.

Temporal and parameter precision requirements are also increased where small deviations materially affect mission outcomes. Result

The original simulation is not treated as automatically sufficient merely because the Simulation Object remains the same.

The changed purpose creates a new Simulation Depth & Precision requirement proportionate to the new decision context.

Architectural Position

Simulation Depth & Precision Module (SDPM) is the fifth and final internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The complete SPSS progression is:

Simulation Purpose → Simulation Scope → Simulation Exclusions → Simulation Applicability Context → Simulation Depth & Precision

Together, these five modules establish:

why the simulation exists,

what it must examine,

what it does not cover,

where and under which conditions it is intended to apply,

and:

how deeply and precisely the inquiry must operate.

SDPM ends the Simulation Purpose & Scope governance space.

The architecture can then proceed to the Reality Representation Standard (RRS), which governs which aspects of reality must be represented inside the simulation and at what abstraction, fidelity, resolution, and limitation.

SDPM does not govern general Validation Depth under VALIDOS®.

The distinction is:

VALIDOS® Validation Depth: How strong and extensive must the validation inquiry be?

SIMULOS® Simulation Depth: How deeply must the simulation inquiry examine the simulated problem?

SDPM also does not govern actual Operational Reality under ORA™, general integrity under INTEGROS®, or computational efficiency itself.

Its governed object is specifically:

the required depth and precision of the simulation inquiry.

Foundational Principle

A simulation should be neither shallower than its purpose permits nor more precise than its underlying simulation basis can legitimately support; depth and precision must remain proportional to what the simulation is intended to determine.

Canonical Closing Statement

Simulation Depth & Precision Module (SDPM) completes the Simulation Purpose & Scope governance layer by establishing how deeply and precisely a simulation inquiry must examine the Simulation Object within its governed purpose, scope, exclusions, applicability context, and consequence environment. Through differentiated depth and precision requirements, materiality, temporal and spatial granularity, parameter and state resolution, scenario and failure depth, uncertainty treatment, sufficiency assessment, anti-inflation rules, proportionality, change governance, and lifecycle traceability, SDPM prevents simulation complexity, computational scale, or apparent numerical precision from being mistaken for methodological adequacy. By requiring simulation rigor to remain proportional to the actual inquiry, SDPM establishes the final governance condition necessary before SIMULOS® proceeds from defining the simulation question to governing how reality itself is represented within the simulation.

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